EC3213 — Money, Credit and Banking

Winter 2022 Examination · Paper 1 · Full Marking Scheme

Module	EC3213 — Money, Credit and Banking
Internal Examiner	Ms. Aisling Finucane
Institution	University College Cork
Session	Winter 2022, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt **2 of 3** (100% of the paper). All questions carry equal weight.
Materials	Non-programmable calculator permitted

Coverage note. This scheme answers **all three** Section A questions in full. A1 and A2 on this paper are worded essentially identically to the Winter 2023 EC3213 sitting — the underlying content below is word-for-word consistent with that scheme. A3 is a new, different 100-mark essay question.

Own-figure and essay-length note. A3 is worth the full 100 marks on its own — budget your exam time accordingly if you plan to answer it.

Source basis. A3's discussion of monetary policy's limitations draws on the module's own material on the zero lower bound, central bank independence and political pressure, and the conflict among monetary policy goals, plus the specific empirical detail on negative interest rate policy's practical constraints (the ECB's policy rate never fell below -40 basis points) and forward guidance's credibility limits, both drawn from the module's own assigned research material.

Section A · Question 1

The market for reserves, the discount rate/IOR corridor, and the money multiplier — 100 marks

A1(a) **[40 marks]**

Explain, with the aid of a diagram, how the Federal Funds interest rate is affected by an Open Market Sale by the Federal Reserve Bank. Illustrate how Open Market Operations are an effective tool for targeting the Fed Funds Rate.

A1(b) **[40 marks]**

Explain with the aid of a diagram how the Discount rate and the Interest rate on excess reserves minimise fluctuations in the Fed Funds rate due to changes in the demand for reserves.

A1(c) **[20 marks]**

Derive and explain the money multiplier.

FULL MODEL ANSWER

The market for reserves: the federal funds rate is the price that clears the **market for bank reserves** — determined by the demand banks have for holding reserves (partly to meet requirements, partly as a cushion against unexpected deposit outflows) and the supply of reserves the Fed makes available.

The demand curve: downward-sloping in the fed funds rate — as i_{ff} falls, the opportunity cost of holding excess reserves (relative to lending them out) falls, so banks want to hold more of them. Since 2008 the Fed has paid interest on reserves (i_{or}) at a fixed spread below the fed funds target, so the demand curve **flattens out (becomes infinitely elastic) once i_{ff} falls to i_{or}** — below that point, banks would rather hold reserves than lend at a rate no better than what the Fed itself pays.

The supply curve: made up of non-borrowed reserves (supplied via OMOs) plus borrowed reserves (banks borrowing directly from the Fed at the discount rate, i_d). If $i_{ff} < i_d$, no bank will borrow from the Fed when cheaper funding is available in the interbank market, so borrowed reserves are zero and supply is **vertical**. If i_{ff} rises above i_d, banks profitably borrow from the Fed and re-lend at i_{ff}, so supply becomes **horizontal (perfectly elastic) at i_d**.

The Market for Reserves: Effect of an Open Market Sale

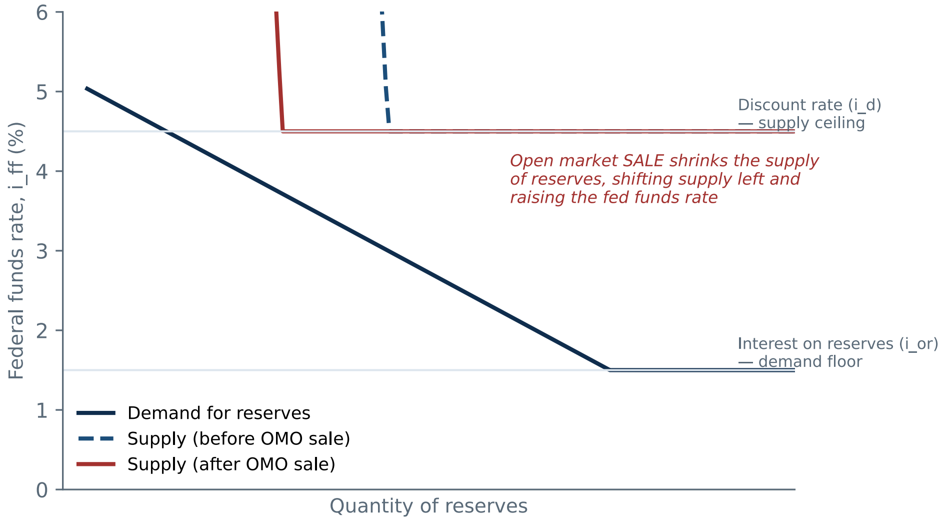

Fig. — An open market sale shrinks the supply of reserves, shifting the (vertical section of the) supply curve left. Where the new equilibrium remains on the downward-sloped section of the demand curve, this raises the federal funds rate from A to B.

> **Effect of an open market sale:** an OMO sale drains reserves from the banking system, shifting the vertical portion of the supply curve to the **left**. Provided the new intersection with demand still falls on the downward-sloped section of the demand curve (i.e. the fed funds rate is still above i_or), this **raises the federal funds rate**. Conversely, an open market purchase shifts supply right and lowers i_ff.

> **Why OMOs are an effective targeting tool:** OMOs give the Fed **complete control over the volume** transacted (unlike discount lending, where the bank itself decides whether to borrow), are **flexible and precise** — any desired quantity of reserves can be added or drained — and can be **quickly implemented and easily reversed** if conditions change, which is precisely what makes OMOs the Fed's primary conventional tool rather than the discount rate or reserve requirements.

EXAM TECHNIQUE

A common exam slip is describing the effect of an OMO sale without specifying *where* the new equilibrium falls relative to i_or — if reserves remain abundant enough that the intersection stays on the flat section of the demand curve at i_or, an OMO sale of typical size has **no effect** on the fed funds rate at all.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31–40	**31–40**	Demand and supply curves in the market for reserves correctly shaped (demand downward-sloping then flat at i_or; supply vertical then flat at i_d) **AND** an OMO sale correctly shown shifting supply left, raising i_ff on the relevant section **AND** at least two specific reasons given for why OMOs are an effective targeting tool.
21–30	**21–30**	Correct diagram shape and correct direction of the OMO sale's effect, but the reasons OMOs are an effective tool are generic or missing.
11–20	**11–20**	Diagram shape roughly correct but the flat sections (at i_or and i_d) are missing or mislabelled, or the direction of the OMO sale's effect is reversed.
0–10	**0–10**	No credible diagram or explanation given.

> **Framing — a different question from the OMO-sale case:** one question asks how a **deliberate policy action** (an OMO sale) moves i_ff by shifting

supply. This is a different question: how the fed funds rate stays **relatively stable** even when the demand for reserves shifts on its own — due to changing bank behaviour (e.g. heightened precautionary demand during a period of stress) — *without* any deliberate policy action at all.

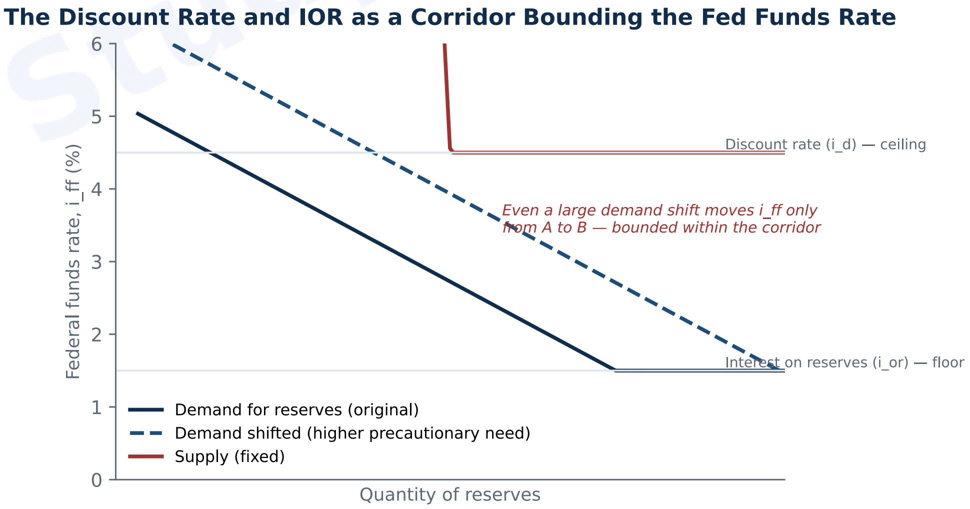

Fig. — The discount rate (ceiling) and interest on reserves (floor) bound the fed funds rate within a corridor. A shift in the demand for reserves that would otherwise move i_ff substantially instead moves it only from A to B, since the curve flattens at each boundary.

How the interest-on-reserves floor works: if a shift in reserve demand (e.g. a fall in banks' desire to hold reserves) would otherwise push the equilibrium fed funds rate below i_or, it can't — because the demand curve itself goes flat at i_or: no bank will lend reserves overnight at a rate below what the Fed itself pays for holding them. The floor is therefore built directly into the shape of the demand curve.

How the discount rate ceiling works: if a shift in reserve demand (e.g. a rise in banks' precautionary demand for reserves, perhaps during a period of financial stress) would otherwise push the equilibrium fed funds rate above i_d, it can't — because the supply curve itself goes flat at i_d: once i_ff would exceed the discount rate, banks simply borrow directly from the Fed's discount window instead of bidding up the interbank rate further, capping i_ff at i_d.

The combined effect — minimising fluctuations: together, the two boundaries form a **corridor (or "channel") system**: as long as the fed funds rate target sits comfortably between i_or and i_d, demand-side shocks to the reserves market — of a kind the Fed did not initiate and may not even have anticipated — are automatically absorbed by banks either lending excess reserves at no less than i_or or borrowing shortfalls at no more than i_d, rather than being fully reflected in a volatile fed funds rate.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31–40	**31–40**	Correctly distinguishes this from an OMO-driven shift (Part (a)) as a demand-side, not supply-side, disturbance **AND** correctly explains both the i_or floor and i_d ceiling mechanisms **AND** the diagram correctly shows a demand shift bounded within the corridor, producing a smaller i_ff change than an unbounded shift would.
21–30	**21–30**	Both the floor and ceiling mechanisms correctly explained, but the diagram doesn't clearly show the demand shift being bounded, or the distinction from Part (a) isn't made.
11–20	**11–20**	Only one of the two boundaries (floor or ceiling) correctly explained.

0-10	**0-10**	No credible explanation of either boundary given.

Setting up the derivation: the monetary base MB is the sum of currency in circulation (C) and total reserves (R) held by banks: MB = C + R. The narrow money supply M1 is currency plus checkable deposits (D): M1 = C + D. Reserves themselves split into required reserves (a fraction, rr, of deposits) and any excess reserves banks choose to hold: R = (rr × D) + ER.

Substituting through: MB = C + (rr × D) + ER. Dividing the M1 definition by the MB definition, and dividing every term by D, lets us express both C and ER relative to deposits — define c = C/D (the public's currency-to-deposit ratio) and e = ER/D (banks' excess-reserve ratio). This gives:

Result: $m = M1/MB = (C+D)/(C + rr \cdot D + ER) = (c+1) / (c + rr + e)$

The simple special case: if we assume away currency holdings and excess reserves entirely (c = 0, e = 0) — i.e. all money stays in the banking system as deposits, and banks lend out every reserve dollar not legally required to be held — the formula collapses to the **simple money multiplier**: m = 1/rr. This is the same result reached by summing the geometric series of successive lending rounds directly.

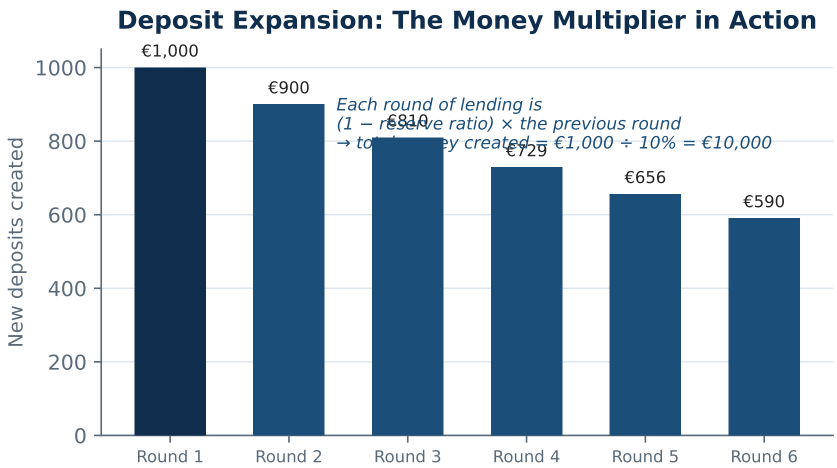

Fig. — The geometric-series intuition behind m = 1/rr: successive rounds of deposit expansion (illustrative: €1,000 initial deposit, 10% reserve ratio), summing to €1,000 ÷ 10% = €10,000.

Round	New deposit created (€)	Held as reserves (€)	Re-lent (€)
1	1,000.00	100.00	900.00
2	900.00	90.00	810.00
...	**(geometric series continues)**		
Total money created	**€10,000.00**		**= €1,000 ÷ 10%**

Why the full derivation matters beyond the simple case: in reality c and e are both strictly positive — the public holds some currency outside banks, and banks hold some excess reserves as a cushion — so the **actual money multiplier m = (c+1)/(c+rr+e) is always smaller than the simple 1/rr formula** would suggest. This matters directly for central bank policy: it's part of why the enormous expansion of the monetary base during quantitative easing (see the conventional/unconventional tools discussion) did not translate one-for-one into a proportional expansion of the broader money supply — c and e both rose sharply during and after the 2008 crisis as both the public and banks became more cautious.

Mark Allocation — Part (c) (20 marks)

Tier	Marks	What separates this tier
15–20	15–20	Correctly derives m = (1+c)/(c+rr+e) from the MB and M1 balance-sheet definitions **AND** correctly shows the simple case m = 1/rr **AND** gives a correct worked numeric example.
10–14	10–14	Correctly states the simple multiplier m = 1/rr with a worked example, but the fuller algebraic derivation (via c and e) is missing or incomplete.
5–9	5–9	Vague reference to "banks lending out deposits" without a reserve ratio, multiplier formula, or derivation.
0–4	0–4	No credible derivation or explanation given.

Question A1 Total: 100 Marks

Section A · Question 2

The Phillips curve family, the Taylor Rule, and theories of money demand — 100 marks

A2(a) [50 marks]

Outline the economic reasoning behind: (a) The Phillips Curve; (b) The Expectations Augmented Phillips Curve; (c) The Accelerationist Phillips Curve; and (d) The Taylor Rule.

A2(b) [50 marks]

Compare the Classical Theory of Money Demand, the Keynesian Theory of Money Demand and Milton Friedman's Theory of Money Demand. In your opinion, which theory (theories) of money demand is most appropriate to describe the current economic environment, why?

FULL MODEL ANSWER

(a) The (traditional) Phillips Curve: A.W. Phillips's 1958 LSE study documented a **negative statistical relationship** between wage inflation and unemployment in the UK — tight labour markets were associated with higher wage inflation, and vice versa; Solow and Samuelson (1960) replicated the finding for the US using price inflation. The mechanism: lower unemployment increases workers' bargaining power, leading to higher nominal wage demands that feed through into higher prices — a wage-price spiral. Read as a stable, exploitable **policy trade-off**, this was the dominant view until Friedman's 1968 critique.

(b) The Expectations-Augmented Phillips Curve: Friedman's key insight was that workers bargain over **real**, not nominal, wages. In a tight labour market, workers would demand nominal wage growth in excess of whatever price inflation they *expect* — giving $\pi_t = \pi_t^e - \gamma(U_t - U^*)$, where expected inflation (π_t^e) now enters the relationship explicitly, but its own formation is left **unspecified** at this stage — a distinct, intermediate step from the accelerationist version below.

(c) The Accelerationist Phillips Curve: Friedman went further and reasoned specifically about how inflationary expectations are determined, arguing they are **adaptive** — this year's expected inflation is simply last year's actual inflation, $\pi_t^e = \pi_{t-1}$. Substituting this specific assumption into the expectations-augmented curve from (b) gives $\pi_t = \pi_{t-1} - \gamma(U_t - U^*)$: it is now the *change* in inflation, not its level, that relates to the unemployment gap. This means a central bank *can* hold unemployment below its natural rate U*, but only by accepting **ever-accelerating** inflation.

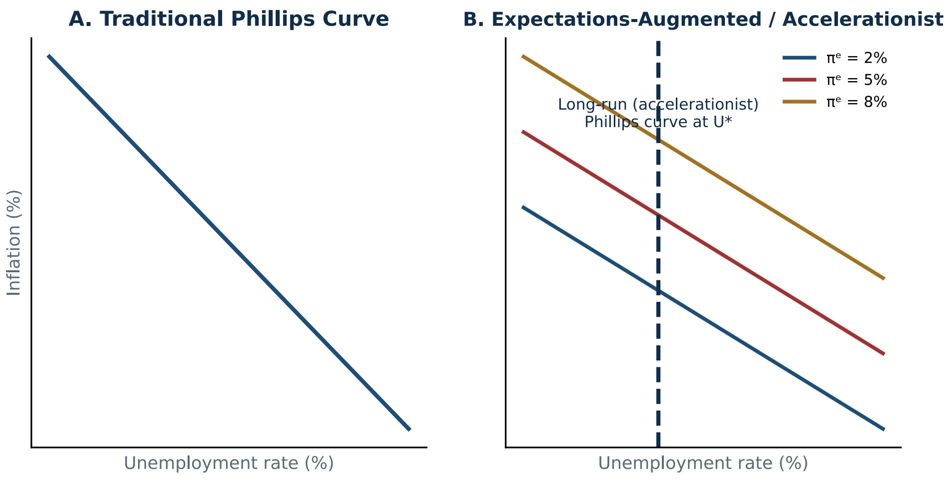

Fig. — The traditional Phillips curve (A) suggests a stable trade-off; the expectations-augmented/accelerationist view (B) shows the curve shifting outward as expected inflation rises, with only a vertical long-run curve at the natural rate U.*

Why the progression matters — and why the accelerationist model was vindicated: loose US fiscal and monetary policy in the 1960s initially produced exactly the traditional trade-off, but as the public's expectations adjusted, the trade-off worsened through the late 1960s, and by the 1970s produced **stagflation** — rising inflation *and* rising unemployment together, a positive correlation the original Phillips curve couldn't explain, but which the accelerationist model predicts exactly once expectations are allowed to adapt.

(d) The Taylor Rule: a simple, transparent rule prescribing how a central bank's policy rate should respond to observable economic conditions: $i\_ff = \pi + r^* + ½(\pi-\pi^*) + ½(\text{output gap})$. Including *both* an inflation gap and an output gap reflects that a central bank cares not only about keeping inflation near target, but also about smoothing business-cycle fluctuations of output around its potential.

EXAM TECHNIQUE

The single most common way to lose marks on this question is treating (b) and (c) as the same thing. They are not: (b) establishes that expectations enter the Phillips curve at all (a general point, compatible with many different ways expectations might be formed); (c) commits to a *specific* mechanism — adaptive expectations — that turns the general expectations-augmented curve into the specific accelerationist prediction.

Mark Allocation — Part (a) (50 marks)

Tier	Marks	What separates this tier
38–50	**38–50**	All four correctly explained **AND** the expectations-augmented curve (b) is clearly distinguished from the accelerationist curve (c) via the explicit adaptive-expectations assumption **AND** the historical stagflation evidence is used to justify the progression from (a) to (c).
25–37	**25–37**	All four named with broadly correct formulas, but (b) and (c) are conflated or not clearly distinguished.
13–24	**13–24**	Only two or three of the four correctly explained.
0–12	**0–12**	Only one, or none, correctly explained.

Classical Theory of Money Demand (Fisher's Quantity Theory): built from the equation of exchange, MV = PY. Fisher assumed **velocity (V) is fairly constant** in the short run (a matter of fixed institutional payment habits) and that output (Y) sits at its full-employment level given flexible wages and prices. Money demand is then Md = (1/V)PY = kPY — a fixed fraction of nominal income, with **no role for interest rates** at all. This makes the theory a clean theory of inflation: with V and Y essentially fixed, $\%\Delta M \approx \%\Delta P$.

Keynesian Liquidity Preference Theory: Keynes identified three motives for holding money — **transactions** and **precautionary** motives (both roughly proportional to income, as in the classical view) and, crucially, a **speculative** motive: money held as a store of wealth, as an alternative to holding bonds, when expected bond returns look unattractive. This makes money demand Md/P = f(i, Y) — negatively related to the interest rate as well as positively related to income — which means **velocity is not constant**: it moves procyclically with interest rates and with changing expectations about future "normal" interest rate levels, undermining the classical link between the money supply and nominal spending.

Friedman's Modern Quantity Theory: Friedman treated money as just another asset in a portfolio-choice framework, with demand for *real* money balances driven by **permanent income** (a longer-run, smoothed measure of wealth, rather than current income) and the expected returns on the alternative assets households could hold instead — bonds, equities, and physical goods (via expected inflation) — each relative to the return on money itself. Friedman argued that because these assets are not close substitutes for money in most portfolios, money demand is only **weakly sensitive to interest rates** — closer to the classical view in its practical stability, but grounded in explicit portfolio-choice reasoning rather than a fixed institutional velocity assumption. This is why Friedman's theory implies velocity is *not* literally constant (as Fisher assumed) but is **stable and predictable** — a crucial practical distinction from the Keynesian view, where velocity's sensitivity to volatile interest-rate expectations makes it genuinely unpredictable.

Comparing Money Demand Theories

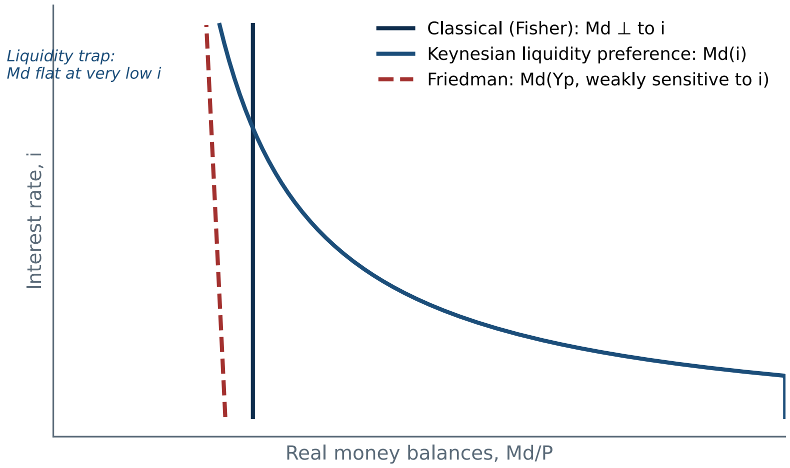

Fig. — Comparing the three theories: Classical money demand is vertical (interest-rate-independent); Keynesian liquidity preference slopes down, including a flat "liquidity trap" region at very low rates; Friedman's is only weakly interest-sensitive, positioned by permanent income rather than current income.

	Classical (Fisher)	Keynesian (liquidity preference)	Friedman (modern quantity theory)
Interest-rate sensitivity	None	Strong (negative)	Weak
Income measure	Current nominal income	Current income	Permanent income
Velocity	Constant	Unstable/unpredictable	Stable, though not constant
Policy implication	Money supply tightly linked to nominal spending	Money supply loosely, unreliably linked to spending	Money supply reliably, though not mechanically, linked to spending

Opinion: which theory best fits the current economic environment?

Assessment: the most defensible position is that **Friedman's modern quantity theory offers the more useful practical framework today**, but not because interest rates are irrelevant — rather because the Keynesian liquidity preference framework's core policy implication (that velocity is so unstable the money supply is barely worth tracking) is less useful in an environment where most major central banks explicitly target an **interest rate** (not a monetary aggregate) as their primary instrument. In that sense, elements of *both* Keynesian and Friedman's reasoning show up in how central banks actually operate: interest-rate targeting (the Keynesian-consistent practical response to unstable velocity) implemented with an eye toward the same permanent-income-driven, portfolio-based stability Friedman emphasised, rather than the classical view's simple, mechanical MV=PY link, which fits poorly with a world of quantitative easing (see Q. A3) where enormous expansions of the monetary base did not translate one-for-one into proportional inflation.

EXAM TECHNIQUE

This is a genuinely open question — a well-reasoned case for the Keynesian view or the classical view can score equally well, provided the argument connects the theory's own assumptions to specific, observable current conditions rather than simply asserting a preference.

Mark Allocation — Part (b) (50 marks)

Tier	Marks	What separates this tier
38–50	**38–50**	All three theories correctly explained with their core mechanisms **AND** a clear opinion given **AND** the opinion is justified by connecting the theory's own assumptions to specific, observable current conditions.
25–37	**25–37**	All three theories correctly explained, and an opinion given, but the justification is generic or doesn't engage with current conditions specifically.
13–24	**13–24**	Only two of the three theories correctly explained, or all three explained with no opinion/justification given at all.
0–12	**0–12**	Fewer than two theories correctly explained.

Question A2 Total: 100 Marks

Section A · Question 3

"Central banks are not all-powerful": the limits of the monetary policy toolkit — 100 marks

A3(a) **[100 marks]**

"Central banks are not all powerful, and their monetary toolkit has only limited applicability." Based on material covered in class and your independent research, discuss the above statement in relation to monetary policy.

FULL MODEL ANSWER

EXAM TECHNIQUE

Like A3 on other EC3213 sittings, this is a single 100-mark essay — commit your full answer time to it if you choose this question. The structure below works through four genuinely distinct types of limitation (a technical floor on rates, an imperfect and lagged transmission chain, external political constraints, and a fundamental one-instrument-many-goals problem), since a strong answer shows the statement is true for several structurally different reasons, not just one repeated point.

Agreeing with the statement — four distinct kinds of limitation

1. A hard technical floor: the zero lower bound. conventional monetary policy works by moving a short-term policy rate — but that rate cannot be cut indefinitely. Once it approaches zero, the central bank runs out of conventional room to stimulate the economy further (see the conventional/unconventional tools discussion elsewhere on this paper), which is exactly what necessitated quantitative easing and forward guidance after 2008. Critically, even these

unconventional tools have their own, separate limits — they are not a costless workaround for the zero lower bound:

- **Negative interest rate policy (NIRP) is politically and socially constrained in practice:** the module's own assigned research notes that no central bank has pushed policy rates deeply negative — the ECB's lowest policy rate reached only −0.40% — because materially negative rates on retail deposits are politically unpopular and can distort bank profitability and behaviour in unpredictable ways. The same research finds that to achieve a given amount of economic stimulus, the policy rate must be cut roughly **twice as much** under NIRP conditions as in normal times — meaning even where NIRP is used, it delivers less stimulus per percentage point than conventional cuts would.
- **Forward guidance's effectiveness depends on credibility the central bank may not have:** forward guidance is, at bottom, a verbal promise about the future path of policy — and the same research shows its stimulative effect depends heavily on how credible the private sector judges that promise to be. Where credibility is low, forward guidance risks being interpreted as "cheap talk" with little real economic effect — a tool whose power depends on a reputational asset the central bank must have already earned, not something it can simply deploy at will.

> **2. An imperfect, indirect, and lagged transmission chain.** a central bank does not control its ultimate goals — price stability, full employment — directly. It can only move its tools, and rely on a chain running from tools through policy instruments and intermediate targets before finally, imperfectly, and only after "long and variable lags," reaching the actual goals (see Q. A1 of the Winter 2025 EC3213 scheme for the full chain). Each link in that chain is itself imperfect and only partially predictable, meaning the central bank's genuine control weakens with every step — it is, at best, steering a system with a long and uncertain delay between action and outcome, not directly setting the outcome itself.

> **3. External political constraints on how far the toolkit can actually be used.** even where a central bank has the formal, legal tools to act — for example, raising rates far enough to bring down persistent inflation — doing so has real costs (slower growth, job losses, higher borrowing costs for households and governments) that provoke political backlash. The module's own material notes explicitly that when a central bank raises rates enough to upset the public, businesses, and politicians, this can lead directly to **calls to limit its independence or reduce its power** — meaning the central bank's practical freedom to use its toolkit fully is itself constrained by the political tolerance for the toolkit's side effects, not just by the toolkit's technical scope.

> **4. A fundamental problem of one instrument, many goals.** a central bank is typically pursuing several goals simultaneously — price stability, high employment, interest-rate and exchange-rate stability, financial stability — but has, in practice, essentially **one primary instrument** (the policy rate) to pursue them all. The module's own material is explicit that these goals can directly **conflict** in the short run: price stability calls for raising rates when the economy is overheating, but doing so risks pushing unemployment up in the short run, forcing the central bank into a genuine trade-off rather than a solution that achieves every goal at once. This is a structural, not merely practical, limitation — no amount of skill or independence lets one instrument hit several distinct targets simultaneously and perfectly.

What the toolkit genuinely cannot reach at all

> **Limited applicability, beyond limited power:** the statement's second half — "limited applicability" — points to something distinct from the four

limitations above: there are entire categories of economic problem monetary policy is simply the wrong tool for, regardless of how skilfully or independently it's wielded. A **supply-side shock** (an oil price spike, a pandemic-driven disruption to global supply chains) raises costs and prices through channels a policy rate cannot reverse — tightening policy to fight the resulting inflation does nothing to fix the underlying supply constraint, and can simply add a demand-side recession on top of an already supply-driven one. Similarly, **structural problems** — a persistent mismatch between the skills the labour market needs and the skills workers have, or genuinely low long-run productivity growth — are not things a central bank's interest-rate lever can fix at all; they call for fiscal, educational, or structural policy instead, and treating them as monetary policy problems misdiagnoses them from the outset.

A balanced conclusion

Agree — central banks are not all-powerful, for several structurally distinct reasons, and their toolkit's applicability is genuinely bounded, not just its raw power.

Summary of the argument: the statement holds up well against the module's own material: a hard technical floor (the zero lower bound) that even unconventional tools only partially work around (both NIRP and forward guidance carry their own separate, real-world limits); an indirect, lagged transmission mechanism that weakens control at every step; genuine political constraints on how far an independent central bank can actually push an unpopular but technically available tool; and a structural one-instrument, multiple-goals problem that has no purely technical solution. Layered on top of all this, some economic problems — supply shocks, structural mismatches — sit entirely outside what any monetary policy toolkit, however skilfully deployed, can be expected to fix. A student arguing that central banks retain substantial effective power despite these limits (e.g. citing successful inflation-targeting track records, discussed in the module's own material) can score equally well, provided the argument directly engages with, rather than ignores, the specific limitations above rather than simply asserting central bank competence in general terms.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
76–100	**76–100**	At least three structurally distinct types of limitation correctly identified and explained (zero lower bound/unconventional-tool limits, transmission-lag imperfection, political constraints, one-instrument-many-goals) **AND** the "limited applicability" half of the statement addressed separately from "not all powerful" (i.e. problems monetary policy is the wrong tool for entirely, not just problems it handles imperfectly) **AND** a clear, well-defended conclusion given.
51–75	**51–75**	At least two types of limitation correctly explained with reasonable depth, and a conclusion given, but the "limited applicability" point (wrong tool entirely, vs. imperfect tool) is not clearly distinguished from general limitations on power.
26–50	**26–50**	Discusses monetary policy limitations in generic terms (e.g. "there are lags") without engaging specific mechanisms from the module's own material.
0–25	**0–25**	No credible discussion of monetary policy's limitations given.

Question A3 Total: 100 Marks